MICHAEL FRYER



Data Scientist & Software Engineer | M.Sc. Applied Data Science Candidate

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Aufenthaltstitel Holder | English (Native) | German (A2)
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SUMMARY

Software engineer transitioning into data science, currently pursuing an M.Sc. in Applied Data Science at Frankfurt School of Finance & Management. Experienced in Python-based machine learning, statistical analysis, and data visualization, with a professional background in full-stack software development. Seeking to apply analytical and engineering skills to data-driven roles in industry.

EDUCATION

M.Sc. in Applied Data Science
September 2025—Present
Frankfurt School of Finance & Management
Frankfurt am Main, Germany
Expected graduation: Summer 2027 | Relevant coursework: Machine Learning 1 & 2, Deep Learning, NLP, Computational Statistics & Probability, Databases & Cloud Computing

B.Sc. in Software Engineering and B.Sc. in Computer Science
August 2015—May 2019
Montana Tech
Butte, MT, USA

EXPERIENCE

Quantitative Analyst - Intern
August 2026—Present
Deutsche Börse Group
Eschborn, Germany
- Developing novel business solutions for the Market Data and Services division, focusing on speed and reliability of data delivery to clients.

Associate Game Programmer
October 2019—July 2021
Heavy Iron Studios
Manhattan Beach, CA, USA
- Worked closely with artists to transform their UI mock-ups into a functional user experience.
- Extended Unreal Engine's replication graph to meet our network performance requirements.
- Shipped production-ready features across multiple Unreal Engine 4 patch releases in preparation for a full commercial game launch.
- Built and deployed the Live Operations portal using Django and AWS, enabling real-time content management for a live-service title post-launch.

Software Engineer
September 2018—August 2019
Butte Food Bank
Butte, MT, USA
- Co-developed FRED in a three-person team, a full-stack inventory and reporting system that streamlined operations for the Butte Food Bank.
- Developed the front-end application of FRED using WPF Forms.
- As sole contractor after graduation, expanded reporting features to help the Food Bank meet government regulations.

PROJECTS

Urban Mobility Risk Analysis *| Python, Pandas, DBSCAN, Streamlit*
Applied DBSCAN clustering to 2.4M geolocated injury accidents across Germany (2016–2025) to identify and rank high-risk zones in any of its 401 districts. Built a public Streamlit dashboard with a district picker, outlier filtering and per-district parameter tuning by binary search.

Agentic Connections *| Python, Anthropic API, pytest*
Red-teaming A/B experiment on a tool-use LLM agent that plays NYT Connections and reports its own results, checked against the game engine's ground truth. 300 pre-registered runs across two Claude models, analyzed with Fisher's exact tests; offline test suite against a fake API client.

TECHNICAL SKILLS

Machine Learning | Neural Networks | Feature Engineering | Data Analysis | Statistical Analysis | Data Visualization | EDA | Python | PyTorch | Scikit-learn | Pandas | NumPy | SciPy | SQL | Plotly | Jupyter | Claude Code | AWS | Docker | Linux/Bash | Git | TypeScript | C/C++

Analytical Thinking | Problem Solving | Attention to Detail | Communication | Collaboration